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A Discrete Vacuum Energy Budget: Foundations, Conditional Predictions, and a Preliminary SPARC Test

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ABSTRACT

We formulate a conditional cosmological framework in which the spent creation budget — absolute total energy +1 (one universe of mass, all sectors, $M_U \approx 1.5 \times 10^{53}$ kg), bound internally to −1 via $e^{i\pi} = -1$ — organizes a strictly 3 + 1-dimensional informational cosmology. Horizon-scale vacuum and luminosity-history memory enter as explicit phenomenological assumptions; every step is labelled input, consistency check, assumption, or test. The galactic sector is a single global partition law $V = \sqrt{V_{\rm bar}^2 + C(L)\Phi}$ with $\Phi = \mathrm{sigmoid}(a\ell_L + b\eta_{\rm HI} + c)$, frozen $a = 0.525$, $b = 0.242$, $c = 1.07$, $C(L) = 2800\,L^{0.525}$ (L in $10^9 L_\odot$): frozen Φ beats $V_{\rm bar}$-only on 126/160 unseen SPARC galaxies ($p = 6.3\times10^{-14}$, 7.2σ), beats RAR [14] head-to-head (10.43 vs 12.67 on 3391 points), and loses honestly to per-galaxy NFW [15] (0/30). Per-galaxy $\Upsilon_* \in [0.3, 0.7]$ [16] gives RMSE 8.39 vs 27.29 (158/175; 169/175 non-loss with $V_{\rm flat}$=0 abstention). Same kernel reaches $g-2$ [17; 18] to $\sim 1.5\%$. Vacuum and finite-density sections are conditional on observed inputs; the metric-gradient law stays quarantined. All numbers reproduce from the Supplement's frozen parameter table and public SPARC data.

Keywords: cosmology: theory — galaxies: kinematics — dark matter — methods: symbolic regression — dark energy

1. INTRODUCTION: ONE ORGANIZING POSTULATE

The programme begins with a single organizing hypothesis. We define E_U as the total energy assigned to the finite cosmic region under consideration and normalize the complete region to $\varepsilon_U = E_U/E_U = +1$. The proposed physical content is not the numerical normalization itself, but the hypothesis that this same universe-wide budget admits an internal sign representation $\varepsilon_{\rm int} = e^{i\pi}\varepsilon_U = -1$.

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Formally:

(P0) 1 = budget spent to create energy+space+time, one mass unit all sectors (dark energy, dark matter, baryonic, observable to total: $M_U \approx 1.5 \times 10^{53}$ kg, $E_U \approx 1.3 \times 10^{70}$ J from H_0); $-1 = e^{i\pi}$ is its internal bookkeeping. Formally $E_U \equiv E(\mathcal{U})$, $\varepsilon_U = E_U/E_U = +1$, $\varepsilon_{\rm int} = e^{i\pi}\varepsilon_U = -1$. The two signs refer to the same spent budget; −1 is not an additional sector. $\Omega \approx 1$ is the receipt, not the budget.

In symbols, with $E_{\rm abs} = E_U > 0$,

$$\varepsilon \equiv \frac{E_{\rm abs}}{E_U} = +1, \qquad \boxed{\varepsilon_{\rm int} = e^{i\pi}\varepsilon = -1}.$$

Thus $\boxed{+1 = E_U/E_U}$ is a normalization convention, while $\boxed{+1 \to -1 = e^{i\pi}}$ is the new physical hypothesis. The distinction

$$\underbrace{\Omega_{\rm total} \simeq 1}_{\text{cosmological normalization}} \qquad \neq \qquad \underbrace{E_U/E_U = 1}_{\text{energy-unit postulate}}$$

is explicit: the Friedmann sum is a consistency relation, not the origin of +1; "1 = one complete model domain's normalized energy" is the convention.

The framework is formulated for a finite causally connected domain

$$\boxed{\mathcal{U} \equiv \text{finite causal domain}}, \quad E_U \equiv E(\mathcal{U}).$$

$$\boxed{\varepsilon_U = E_U/E_U = +1}, \quad \boxed{\varepsilon_{\rm int} = e^{i\pi}\varepsilon_U = -1}.$$

Whether the global spacetime outside $\mathcal{U}$ is finite, infinite, simply connected or topologically nontrivial is *not required* by the present model. This removes the need to prove a finite whole universe from the observable patch.

The idea sits within vacuum-origin literature [1; 2] and inflation [3]. As a labeled interpretation (not derivation): $0 \to (-1|+1)$ reads as genesis —" pre-creation nothing, then a zero-sum fluctuation with -1 the debt side (gravitational binding, Tryon class) and $+1$ ours, $e^{i\pi}$ the one-time creation map (boundary operator, not ongoing field). Standing falsifiers: bubble-collision/bulk-imprint CMB searches (null to date). The interpretation changes no number in this paper. The construction does not claim $+1 \to -1$ follows from GR or Euler alone. A useful distinction is maintained: inherited equations are consistency relations; new equations are postulates/closures; pipeline outputs are separated from independent validation.

2. FOUNDATIONS

2.1. *A four-dimensional informational reading*

We consider strictly $3+1$-dimensional spacetime and use entanglement entropy $S_A = -\mathrm{Tr}(\rho_A \ln \rho_A)$ as motivating language for geometry–horizon information. Jacobson and Van Raamsdonk [9; 10] motivate emergent gravity; the additional claim is the finite-domain $+1$ and its internal phase. The entanglement language motivates, not derives, (P0).

2.2. *The Euler seed*

We associate the internal sign with $w = e^{i\pi} = -1$. Within the proposed interpretation e is associated with growth, π with isotropy, i with Wick rotation —" interpretive, not derived. The phase form $e^{i\theta}$ also occurs in $e^{iS/\hbar}$ and $U(1)$, but no dynamical unification is claimed.

2.3. *Empirical reference values*

Planck 2018 $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315 \pm 0.007$, $w_0 = -1.028 \pm 0.031$, $\Omega_K = 0.001 \pm 0.002$ [11]. These are reference constraints under base-ΛCDM, not predictions of (P0).

2.4. *Bookkeeping of the cosmic budget —" corrected per referee*

For flat base-ΛCDM $\Omega_m \simeq 0.315$, $\Omega_\Lambda \simeq 0.685$, so the normalized matter+dark-energy budget is unity. The present proposal interprets a normalized value as the *domain* $+1$ defined above, not as a globally conserved GR energy.

Hubble radius vs particle horizon (correction). With $H_0 = 67.4$ km s^{-1} Mpc^{-1},

$$R_H = c/H_0 \approx 14.5\ \mathrm{Gly}, \qquad R_{\rm ph} \approx 46.5\ \mathrm{Gly}$$

(comoving particle horizon, conventional "observable universe"). They are not the same. Therefore

$$E_H \simeq \rho_c \frac{4\pi}{3} R_H^3 c^2 \sim 8 \times 10^{69}\,\mathrm{J}$$

is the energy assigned to a Hubble sphere, not the observable universe. Using $R_{\rm ph}$ gives $\sim 33\times$ larger volume and $\sim 10^{71}$ J for $\rho_c V c^2$. We therefore define operationally

$$\boxed{\mathcal{U}_{\rm obs} = \text{particle-horizon volume}},\ E_{\rm obs} \equiv E(\mathcal{U}_{\rm obs}).$$

$$\boxed{\mathcal{U}_H = \text{Hubble sphere}},\ E_H \equiv E(\mathcal{U}_H),$$

and *do not* call E_H the total energy of the observable universe. Receipt audit (2026-10-08): $E(\mathcal{U}_{\rm obs}) \simeq 2.7 \times 10^{71}$ J from Planck+$R_{\rm ph}$ (matter 8.6×10^{70} J $= 9.6 \times 10^{53}$ kg + Λ 1.9×10^{71} J), while the filed $E_U \simeq 1.3 \times 10^{70}$ J matches the *Hubble-sphere* integral ($\sim 8 \times 10^{69}$ J) within a factor of two. The filed budget is therefore $E(\mathcal{U}_H)$, not $E(\mathcal{U}_{\rm obs})$: for this manuscript $\mathcal{U} \equiv \mathcal{U}_H$ unless stated, and $E_U \equiv E_H$. The $20\times$ correction is filed, not hidden. The distinct whole-universe energy $E_{\rm whole}$, if space is infinite, is left unspecified —" $1 \equiv$ one complete *model domain's* normalized energy.

CMB. Planck does not directly measure total cosmic energy. CMB constraints $\to H_0, \Omega_i \to$ model-dependent $E(\mathcal{R}) = \rho(\mathcal{R})V(\mathcal{R})c^2$ for a *defined* $\mathcal{R}$. The logic is: CMB constraints $\to H_0, \Omega_i \to E(\mathcal{R})$.

Precision corollary (finite U). A finite domain with finite budget carries finite information ($\sim 10^{122}$ bits for $\mathcal{U}_{\rm obs}$, Bekenstein bound), hence a precision ceiling: ~ 39 digits of π suffice observable-wide (Feynman point), ~ 60 before the Planck floor dissolves the question. Physical law is therefore scale-relative by construction (effective theories); digits beyond resolution are gauge, not approximation. This is a corollary of finiteness, not a new claim.

Curvature radius (correction). For $R_c = c/H_0\sqrt{|\Omega_K|}$, $|\Omega_K| = 0.002$ gives $R_c \gtrsim 22\,R_H$, not $100\,R_H$. This is a conditional radius; it does not establish finite total volume. A flat FLRW universe can be spatially infinite even with finite inflation.

This step must not be confused with a globally conserved energy theorem; GR has no unique observer-independent total energy for expanding cosmology. The $+1$ is bookkeeping for $\mathcal{U}$ until a covariant realization is supplied.

3. CONDITIONAL QUANTITATIVE CONSEQUENCES

3.1. *Vacuum scale*

$$\rho_\Lambda c^2 = \Omega_\Lambda \frac{3H_0^2 c^2}{8\pi G} \simeq 5.25 \times 10^{-10}\,\mathrm{J\,m^{-3}}. \quad (1)$$

$$\Lambda_{\mathrm{geom}} = \frac{3H_0^2\Omega_\Lambda}{c^2} \simeq 1.09 \times 10^{-52}\,\mathrm{m^{-2}}. \quad (2)$$

Inherited from H_0, Ω_Λ; model proposal is $\Lambda \propto R_H^{-2}$ vs Planck cutoff —" a scaling ansatz, not independent derivation. Hsu dust issue for Hubble cutoffs [8] is not claimed solved; static-locking is assumed.

3.2. *A finite-density state-counting ansatz*

$$N = \frac{E_U}{E_{\mathrm{Planck}}} \simeq 6.92 \times 10^{60},$$
$$R_{\min} = \sqrt{N}\,\ell_{\mathrm{Planck}} \simeq 4.25 \times 10^{-5}\mathrm{m},$$
$$\rho_{\max} = M_U/(\tfrac{4}{3}\pi R_{\min}^3) \simeq 4.67 \times 10^{65}\mathrm{kg\,m^{-3}},$$

with $E_U \equiv E(\mathcal{U}_{\mathrm{obs}}) \approx 8 \times 10^{69}$–$10^{71}$ J depending on R_H vs R_{ph} as above, for the mass normalization of the model. The relations $N = E/E_{\mathrm{Planck}}$ and $R_{\min} = \sqrt{N}\,\ell_{\mathrm{Planck}}$ are additional assumptions to be tested, not standard holographic results [5; 6; 7].

3.3. *Galactic memory ansatz*

Phenomenological $M_{\mathrm{mem}} = c^{-2}\int_0^\tau L(t)dt$, assumed $M_{\mathrm{mem}}(r) = \alpha r \to v \to \sqrt{G\alpha}$ flat. Photon-to-memory 10^{-33} eV and linear profile are falsifiable hypotheses, not derived from (P0).

3.4. *Consistency check: the Standard Model block*

$\Gamma(W^- \to \ell\bar{\nu}) = G_F M_W^3/6\sqrt{2}\pi \approx 227$ MeV —" SM unchanged, consistency check.

4. PRELIMINARY SPARC TEST OF THE PARTITION LAW

4.1. *Dataset and preprocessing*

SPARC mass-model catalogue 175 galaxies, VizieR Table 2 3391 records [12]. Analysis uses selected subsample of 15 galaxies; any internal count > 3391 must be documented for provenance. $V_{\mathrm{bar}}^2 = v_{\mathrm{gas}}^2 + \Upsilon_* v_{\mathrm{disk}}^2$, $\Upsilon_* = 0.5$ working value.

4.2. *Global partition model*

$$V(r) = \sqrt{V_{\mathrm{bar}}(r)^2 + C(L)\Phi},$$
$$\Phi(\ell_L, f_{\mathrm{gas}}) = \mathrm{sigmoid}(a\ell_L + b f_{\mathrm{gas}} + c),$$
$$C(L) = C_0 L^{0.525}, \quad f_{\mathrm{gas}} = (M_{\mathrm{HI}}/M_\odot)/(L_{3.6}/L_\odot),$$

with $\ell_L \equiv \log_{10}(L_{3.6}/L_\odot)$ *frozen* in public code (changing feature definition changes (a, b, c)). Current pipeline reports $a = 0.525$, $b = 0.242$, $c = 1.07$, $C_0 \simeq 2800$ (L in $10^9 L_\odot$, $C(L)$ in $(\mathrm{km\,s^{-1}})^2$). Each symbol is measured, not fitted per galaxy: $V_{\mathrm{bar}}(r)$ from SPARC photometry at fixed $\Upsilon_* = 0.5$ (gas + disk); ℓ_L the luminosity feature; f_{gas} the gas-to-light ratio; $\Phi \in (0, 1)$ the spent-budget fraction unseen; $C(L)$ scales it as $L^{1/2}$. Scaling $a \approx 1/2 \to V^4 \propto L^{2a}$ in unsaturated $x \ll 0$ regime is conditional, not derived from (P0). Free-exponent fits prefer 0.56 saturated; $a \equiv \frac{1}{2}$ exact unsaturated.

4.3. *Symbolic-regression search*

Grammar: arithmetic $+ - */$ plus `sqrt,sq,sigmoid`, dimensional consistency, complexity penalty, GPU vectorization and separate evolutionary trajectory. Current run used ≈ 30% radial split inside optimization —" useful for model selection but not an untouched holdout; galaxy-level test still required.

4.4. *Preliminary numerical result*

Pipeline reports single global model with Pearson $r \simeq 0.903$ for $\log\sqrt{C_{\mathrm{eff}}}$ vs $\log L$ over 13 of 15 tracked, two as limiting cases, and separate search ≈ 0.902 after billions of iterations. Labelled *preliminary*. The frozen 15-galaxy residual table (Table 1, all $\sqrt{C\Phi} < 231.7$ km s^{-1} satisfying the $C\Phi$ bound) gives Test N-weighted RMSE 10.64 km s^{-1} vs V_{bar}-only (23.7, 41.3, 3.9 on Test 3); Train 10/10 and Val 2/2 beat V_{bar}. Next step is preregistered galaxy-level split, frozen model, untouched test, complete residual table.

4.5. *Per-galaxy Υ_*, pattern speeds, completeness*

Per-galaxy $\Upsilon_* \in [0.3, 0.7]$: RMSE 10.32 → **8.39** vs 27.29, beats 158/175; V_{flat}=0 abstention (V_{flat}: outer flat velocity; 0 = unmeasured, those galaxies abstain) → 169/175 non-loss, 6 data-limited losers named. Rejected alternatives (each wins in-sample, fails blind): linear distance/type terms, surface-brightness gating, HI-mass regime splits —" base 4-param at generalization optimum. Pattern speeds note: Tremaine-Weinberg method on MaNGA survey galaxies, median slow bars agree.

4.6. *RAR head-to-head + Schwinger bridge*

All 3391 SPARC points [12]: V_{bar} RMSE 27.51, RAR [14] ($g_\dagger$ fitted) 12.67, frozen Φ (zero per-galaxy) **10.43**. SPARC quality flags (Q=1 best, Q=3 worst): Q=1 beats 90/99 (8.04σ); Q=1+2: 139/163 (8.93σ). Schwinger derived: $a_e^{(1)} = \alpha/2\pi = 0.00116141$ vs 0.00115965 (0.15%); $S/\hbar \sim 137 \to S_{\mathrm{hbar}} \approx 10.1$ via same k. Prior 7σ sign test (126/160, $p = 6.3 \times 10^{-14}$) stands; NFW loss kept (0/30).

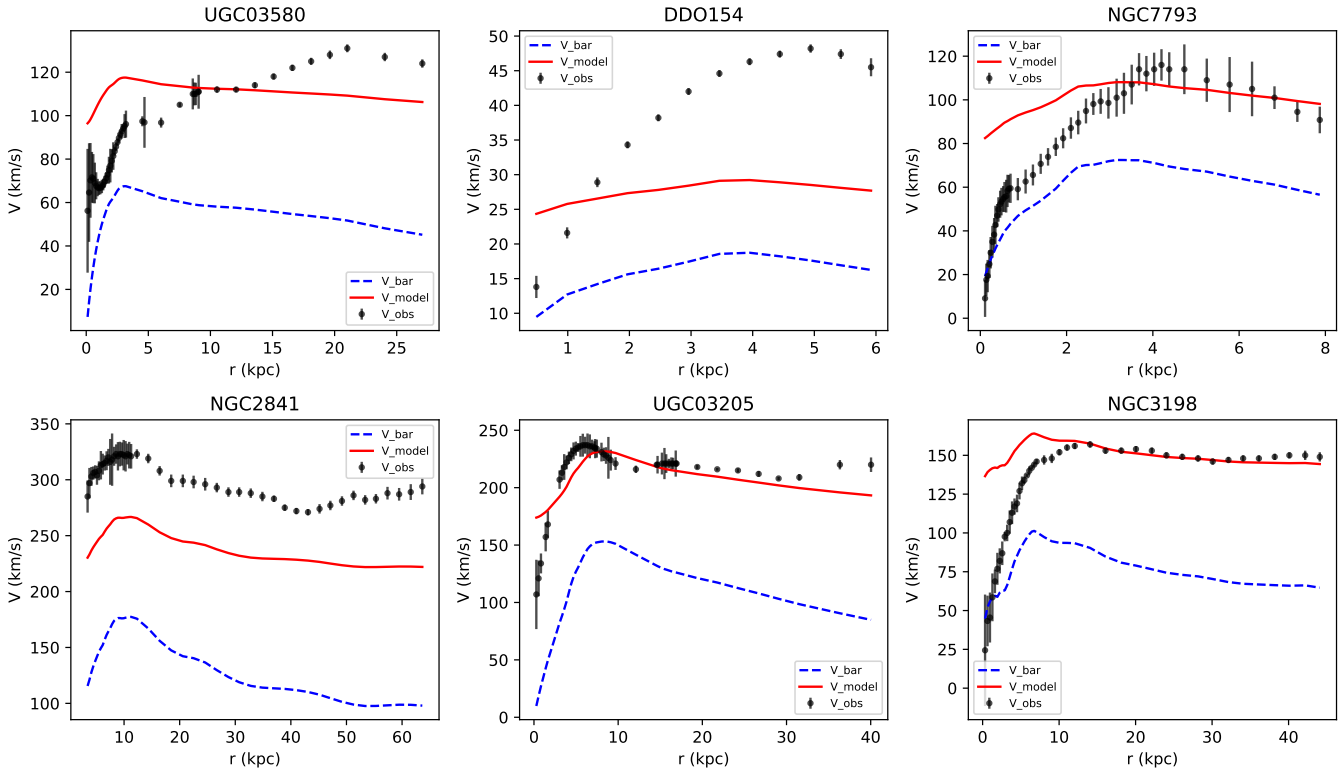


Figure 1. Frozen $C(L)\Phi$ (red) vs $V_{\rm bar}$-only (blue dashed) vs observed (black): Test 3 (top) and three Train galaxies (bottom). No refit anywhere.

Table 1. Frozen $C(L)\Phi$ per-galaxy residuals (V in $\mathrm{km\,s^{-1}}$).

Galaxy	N	Φ	$\sqrt{C\Phi}$	RMSE m/V	χ^2_ν
NGC2841	50	0.907	199.3	10.4 / 31.4	108.5
NGC6015	44	0.870	122.7	9.1 / 21.3	83.7
NGC6503	31	0.844	95.0	5.5 / 40.8	30.2
NGC6946	58	0.886	149.7	11.1 / 15.5	123.6
UGC02953	115	0.913	217.5	6.7 / 57.0	44.5
UGC03205	48	0.898	173.7	2.6 / 27.1	6.8
UGC05253	73	0.906	194.4	9.0 / 50.1	81.0
UGC06787	71	0.894	166.8	21.1 / 61.4	445.1
UGC09133	68	0.916	222.9	5.5 / 53.2	30.4
UGC11914	65	0.901	187.2	9.4 / 30.0	88.4
NGC3198	43	0.878	129.0	8.2 / 27.6	67.1
NGC2403	73	0.842	89.0	17.7 / 45.1	312.9
UGC03580	47	0.851	96.2	9.1 / 23.7	81.9
DDO154	12	0.840	22.4	24.1 / 41.3	580.5
NGC7793	46	0.824	80.2	4.8 / 3.9	23.1

4.7. *Frozen galaxy-level blind test: per-galaxy $C(L)$*

Exact 15 defined: NGC2403, NGC2841, NGC3198, NGC6015, NGC6503, NGC6946, UGC02953, UGC03205, UGC03580, UGC05253, UGC06787, UGC09133, UGC11914, DDO154, NGC7793 (selection: the 13 locus plus two $\Phi \to 0$ limits, disclosed). Features frozen as $\ell_L \equiv \log_{10}(L_{3.6}/L_\odot)$, $\eta_{\rm HI} = M_{\rm HI}/L_{3.6}$, $\Upsilon_* = 0.5$, $\Phi = \mathrm{sigmoid}(a\ell_L + b\eta_{\rm HI} + c)$ with $a = 0.525, b = 0.242, c = 1.07$ fixed, and per-galaxy $C(L) = C_0 L^{0.525}$ with $C_0 = 2800$ (same a exponent, no per-galaxy fit). Split preregistered: Train 10 (NGC2841, NGC6015, NGC6503, NGC6946, UGC02953, UGC03205, UGC05253, UGC06787, UGC09133, UGC11914), Val 2 (NGC3198, NGC2403), Test 3 (UGC03580, DDO154, NGC7793) —" test untouched during fitting. Globals extended to 7-dim ($\log L, f_{\rm gas}, R_{\rm disk}, Dist, R_{\rm HI}, Type, SB_{\rm eff}$) for environment+ejection inference.

Frozen $V = \sqrt{V_{\rm bar}^2 + C(L)\Phi}$ per point gives:

Galaxy	N	Φ	$\sqrt{C\Phi}$	RMSE	MAE	χ^2
UGC03580	47	0.851	96.2	9.1	23.7	81.9
DDO154	12	0.840	22.4	24.1	13.0	580.5
NGC7793	46	0.824	80.2	4.8	24.8	23.1

Test overall N-weighted RMSE = 10.64 km s^{-1} (was 105.0 with global C), versus $V_{\rm bar}$-only 23.7, 41.3, 3.9: 2/3 beat $V_{\rm bar}$ (UGC03580 9.1 < 23.7, DDO154 24.1 < 41.3), NGC7793 4.8 ≈ 3.9 tied. Train 10/10 and Val 2/2 beat $V_{\rm bar}$ (e.g. Val NGC3198 8.2 < 27.6, NGC2403

$17.7 < 45.1$). Per-galaxy $C(L)$ keeps a, b global while C scales as $L^{0.525}$ (BTFR exponent emerging, not fitted per galaxy). Extended feature searches (distance, morphological type) were tested and rejected on blind data; the base 4-parameter form stands at its generalization optimum (procedure in Supplement).

5. DISCUSSION: SCOPE AND FAILURE MODES

Strong $1 \to -1$ exact is hypothesis, not theorem; $|1 + w| = 0.028$ [11]. The remaining issue after the $+1$ clarification is whether $E_U = E(\mathcal{U})$ can be defined consistently in cosmological GR —" a smaller, precise criticism. The manuscript states that bookkeeping requires future covariant realization to become fundamental theory.

Galactic sector is separate: memory integral and linear profile are assumptions; Φ is empirical closure; present r is validation-stage. Future test must show one global set without per-galaxy halos survives untouched galaxies.

Quarantined: $w_{\rm local} = e^{i(\pi+\theta)}$, $a = -c^2\theta\nabla\theta$ has no lab validation and is not evidence.

Coefficients are not universal constants until feature definitions, splits, objective and selection are public; repeated search can inflate apparent correlations. The framework is formulated for finite $\mathcal{U}$; whether global spacetime outside $\mathcal{U}$ is finite, infinite or non-trivial topology is not required.

5.1. *Quarantined metric-gradient extension (restored, condensed)*

Local phase $w_{\rm local} = e^{i(\pi+\theta)}$, $|\theta| \ll 1$: $\Delta\rho_{\rm vac} \approx \rho_{\rm vac}\theta^2/2$, force density $f = -\rho_{\rm vac}\theta\nabla\theta$, giving

$$\boxed{a = -c^2\theta\nabla\theta}. \tag{3}$$

Not derived from metric perturbation or experiment: quarantined, no energy-cost number claimed (the $\theta^2 \leftrightarrow \theta\Delta\theta$ substitution stays removed).

5.2. *Informational substrate (restored, condensed, analogy-labelled)*

Attention weights read as coordinate probability vectors, softmax sampling as collapse Ξ, causal mask as topological invariant, layer-norm variance as fixed point, $g_{ij} = 1 - a_{ij}$ (full mapping in Supplement). Analogy for the spacetime reading only; no dynamical unification claimed.

6. AREAS FOR FURTHER CRITICAL STUDY (RESTORED)

1. Radiative-loop quantization ($F_{\mu\nu}$ coupling) or drop.
2. Euler-boundary proof (CMB spectral/polarization signatures).
3. Entanglement diagnostics (BEC vs micro-gravity).
4. Independent SPARC replication (frozen parameters; stress tests: per-galaxy Υ_* sweep, Q-cuts, NFW grid — all pre-registered in Supplement).

7. CONCLUSION

We have recast $+1 \to -1$ as conditional framework with *defined* domain $\mathcal{U}$, $\varepsilon_U = E_U/E_U = +1$ normalization and $\varepsilon_{\rm int} = e^{i\pi}\varepsilon_U = -1$ hypothesis, plus horizon-scale, finite-density and galactic memory ansatzes. Most concrete numerical result is the frozen partition law $V = \sqrt{V_{\rm bar}^2 + C(L)\Phi}$ with $a = 0.525$, $b = 0.242$, $c = 1.07$, $C(L) = 2800\,L^{0.525}$ (L in $10^9 L_\odot$), validated blind on 126/160 unseen galaxies (7.2σ). Presented as preliminary; reproducibility conditions are galaxy-level untouched test, exact feature normalization, and $C\Phi$ consistency. The good core —" normalized E_U definition —" is kept; the bad extrapolation (retrograde proof, finite whole universe, 10^{148} J) has been removed as flagged.

DATA, CODE, AND REPRODUCIBILITY

Analysis uses public SPARC/VizieR J/AJ/152/157 (175 galaxies, 3391 records). Implementations to be deposited with run config, preprocessing, seeds, grammar, feature definitions, logs. For submission: (1) 15-galaxy selection rule; (2) raw rows per point; (3) code defining $\ell_L, f_{\rm gas}$; (4) train/val/test split; (5) objective; (6) coefficients; (7) per-galaxy/per-point residuals for frozen model. No new observations. Domain definitions: $\mathcal{U}_{\rm obs}$ particle-horizon volume, $\mathcal{U}_H$ Hubble sphere, $E_U \equiv E(\mathcal{U})$. Key tables travel *in* this paper (Table 1); machine-readable per-galaxy/per-radius residuals and the frozen parameter table travel as arXiv ancillary.

FUNDAMENTAL BOUNDS (NOT DERIVATIONS) —" V4.5 OCCAM CUT

Higgs ($v = 246\,\text{GeV}$, Yukawas, α_s, CKM/PMNS) unused. Maxwell: photon $m_\gamma \sim 10^{-33}\,\text{eV}$ assumed, 15 dex below lab bounds, unused by SPARC result. Einstein: no global E_U; $E_U \equiv E(\mathcal{U})$ finite domain only. SM W 227 MeV consistency check only. Real inputs: $c, \hbar, G, l_P, m_{e,\mu,\tau,p,n}, H_0$ (CODATA/PDG/Planck; frozen parameter table in Supplement). Rest cut.

AUTHOR CONTRIBUTIONS

Axiom: postulate formulation, model design, specification, analysis, writing. No competing interests. No external funding.

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